

Assignment Category: CAT\_03

# SportNest - Sports Facility Booking Management System

 [Explanation Video](#) 

## Project Overview and Discussion

You are going to build a full-stack sports facility booking platform using the MERN Stack along with Better Auth Authentication.

The system allows users to explore available sports facilities such as football turfs, badminton courts, swimming lanes, or tennis courts, and make bookings for specific dates and time slots.

This system reflects a real-world sports reservation portal.

## How the System Works

- Users can browse available facilities.
- Authenticated users can book a facility.
- Logged-in users can add and manage their own facilities.
- Users can view or cancel their bookings.
- Facility owners can update or remove facilities.
- Booking data will be stored and managed from the database.

## Similar Platforms for Inspiration

- [Playo](#)
- [BookMyTurf](#)

## Avoid Projects Related To

- Recipe Website
- Job Portal / Marketplace
- Existing Course Module Concepts

## Ensure the Following Things to Get 100% Mark

- **GitHub Commits:**
  - Include a minimum of 15 notable GitHub commits on the client side.
  - Include a minimum of 8 notable GitHub commits on the server side
- **Readme.md:** Include README with:
  - Project Name
  - Purpose
  - Live URL
  - Features
  - NPM Packages Used
- **Lorem Text:** Don't use any Lorem ipsum text; you can not use the default alert to show any error or success message.
- **Host your Application:** You can choose deployment systems like Vercel, Render for hosting. As you develop a single-page application
  - Ensure that the page doesn't throw any error on reloading from any routes.
  - Logged in User must not be redirected to Login on reloading any private route
- Make the site responsive on mobile, tablet and desktop.
- Secure MongoDB credentials using environment variables.
- Design should be clean and recruiter-friendly.
- Do not copy any assignment/module project.

## Deployment Guideline

- Ensure the server does not throw CORS / 404 / 504 issues.
- Live site should work perfectly.

- Reloading any route should not throw errors.
  - Logged-in users must not redirect to login on private route reload.
- 

## **Main Requirements:**

### **1. Layout Structure**

#### **Header/Navbar**

Show:

- Logo + Site Name
- Home
- All Facilities
- My Bookings (Private)
- Add Facility (Private)
- Manage My Facilities (Private)

If logged in:

Show profile dropdown:

- My Bookings
- Add Facility
- Manage My Facilities
- Logout

If not logged in:

Show Login button.

#### **Footer**

Include:

- Contact Information
  - Social Links
  - Copyright
- 

## Home

### Banner Section

Create hero section with:

- Title
- Short description
- Explore Facilities button

### Dynamic Section (Required)

Create Featured Facilities section using database data.

Show a minimum 6 cards.

Each card must contain some details with a Book Now Button.

### Extra Static Sections (Any Two)

Students will need to create two additional sections using their own creativity.

---

## Authentication

### User Login

Create a Login page.

Show title and form.

Fields:

- Email
- Password
- Login Button

If login successful:

Navigate the user to the desired route or Home.

If failed:

Show error using toast or UI message.

Show:

- Register Link
- Google Login Button

Google login should redirect to the desired route or Home.

### **User Registration**

Create a Register page.

Fields:

- Name
- Email
- Photo URL
- Password
- Register Button

If registration succeeds:

Redirect to the Login page.

If fails:

Show error.

Password Validation:

- At least 6 characters
- One uppercase letter
- One lowercase letter

Show:

- Login Link
- Google Login

💡 Don't implement email verification or the forget password method, as it will inconvenience the examiner. If you want, you can add these after receiving the assignment result.

---

## **CRUD Operation**

### **Add Facility (Create)**

Private Route

Create a form and store all data in MongoDB.

Fields:

- Facility Name
- Facility Type
- Image Upload (imgbb/postimage)
- Location
- Price Per Hour
- Capacity
- Available Time Slots
- Description

Auto-fill:

- Owner Email

Store in:

facilities

### **All Facilities (Read)**

Public Route

Show all facilities.

Card should show some information.

If an unauthenticated user clicks Book Now:  
Redirect to Login.

### **Facility Details + Booking**

Private Route

Route:

/facility/:id

Show all facility data.

Booking Form:

- Facility Name
- Booking Date
- Time Slot
- Hours
- Total Price

Save booking in:

bookings

Default:

status: "pending"

### **Update/Delete Facility**

Private Route (**Manage My Facilities route**)

Only facility owner can:

- Update Facility
- Delete Facility

Use confirmation before delete.

## **My Bookings**

Private Route

Show logged-in user bookings.

Display:

- Facility Name
- Booking Date
- Time Slot
- Price
- Status

Action:

- Cancel Booking
- 

## **Other Requirements**

### **Loading Page**

Show loading spinner when data is fetching.

### **Not Found Page**

Create a custom 404 page.



Show:

- Friendly error message
- Back Home button

### UI Design Requirements:

- **Unique Design:** First, decide what kind of website you want to make. Then, search online or check out websites like ThemeForest to get ideas for the design. But remember, your website idea shouldn't be similar to any projects you've done before or to any examples in our modules or conceptual sessions.
  - ★ You can also look for free resources on [blogs](#) to help with your website.
1. Keep the main heading style (font, size, color) consistent across all sections.
  2. Keep paragraph spacing balanced and text easily readable.
  3. Maintain uniform image sizes and spacing.
  4. Use the same button style as on the home page.
  5. Ensure good spacing and proper alignment.
  6. Navbar, Keep the heading/logo same style and size as on the home page.
  7. Use a grid layout with equal image sizes.
  8. Keep all cards equal height and width (especially in services, projects, or products section)
  9. Use the new X logo instead of the old Twitter bird to match the latest rebrand
  10. Responsiveness: Make it responsive for all devices, including mobile, tablet, and desktop views.

Resources:

- <https://uiverse.io/>
- <https://devmeetsdevs.com/>
- <https://bootcamp.uxdesign.cc/free-images-and-resources-collection-for-website-c77f2fc46ce5>

- [https://themeforest.net/?srsltid=AfmBOopTj6PNz51iuV2YJXU+BP8nt19\\_zT5LG2dToAjIHQqzNCzregn0](https://themeforest.net/?srsltid=AfmBOopTj6PNz51iuV2YJXU+BP8nt19_zT5LG2dToAjIHQqzNCzregn0)
  - <https://codecanyon.net/?srsltid=AfmBOooRoUfeK7lOROpchCuA4hPVj5P9WRmtDQJ9K0E6Yhf4VTrHhXKt>
- 

## Challenges

### 1. Search and Filter

Implement search by facility name.

Implement filter by sport type.

Use MongoDB operators:

`$regex`

`$in`

### 2. JWT with Cookies

Implement:

- Generate token
- Store in HTTPOnly cookie
- Verify in middleware
- Protect private APIs

## Optional Requirements

Choose minimum two:

- Theme Toggle
- Framer Motion

## Database Architecture

## Facilities Collection

name  
facility\_type  
location  
price\_per\_hour  
capacity  
available\_slots  
description  
owner\_email  
booking\_count

## Bookings Collection

facility\_id  
user\_email  
booking\_date  
time\_slot  
hours  
total\_price  
status

## What To Submit

- Client-side GitHub Repository
- Server-side GitHub Repository
- Live Website Link